

Practice Problem Set 6

ECON 101 – Macroeconomics

Fall 2025

Problem 1

The Bank of Canada is concerned that the inflation rate (3.8%) is above the 2% inflation-control target midpoint. It decides to implement **contractionary monetary policy**.

- (a) Explain how the Bank of Canada implements contractionary monetary policy using its *key policy interest rate*. Include the roles of the overnight rate target, settlement balances, and open-market operations.
- (b) Describe the complete **monetary transmission mechanism**, step by step, from the initial interest rate change to its effects on real GDP and inflation. Your explanation should include effects on:
 - consumption,
 - investment,
 - the exchange rate,
 - net exports,
 - aggregate demand.
- (c) Using an AD–AS diagram, explain how contractionary monetary policy reduces inflation and brings the economy back toward the 2% target.

Problem 2

The Bank of Canada uses a simplified Taylor-rule style reaction function to guide its target overnight interest rate:

$$i = 1 + 1.5(\pi - 2) + 0.5(Y - Y^*)$$

where:

- i = target overnight interest rate (in percent),
- π = current inflation rate (percent),
- 2 = inflation target midpoint (percent),
- $Y - Y^*$ = output gap (percent).

Suppose new economic data show:

$$\pi = 3.4\%, \quad Y - Y^* = +1.5\%.$$

- (a) Compute the Bank of Canada's recommended overnight interest rate using the reaction function. Show all steps.
- (b) State whether the resulting monetary policy stance is expansionary or contractionary, and explain why.
- (c) Explain how this policy rate affects:
 - bank lending,
 - long-term interest rates,
 - aggregate demand,
 - inflation in the medium run.

End of Problems